
Learning Sparse Visual Representations via Spatial-Semantic Factorization

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Abstract

Self-supervised learning (SSL) faces a tension between semantic understanding and image reconstruction. High-level semantic SSL methods such as DINO encourage transformation-invariant global representations for augmentation alignment, while reconstruction-oriented methods such as MAE preserve dense feature grids for spatial grounding but often yield weaker semantic abstractions. We introduce STELLAR, a framework that alleviates this tension by factorizing visual features into a low-rank product of semantic concepts and their spatial distributions. This disentanglement allows us to perform augmentation alignment on semantic tokens while maintaining spatial localization through a localization matrix for reconstruction. We demonstrate that as few as 16 sparse tokens under this factorized form can simultaneously support high-quality reconstruction (2.60 FID) and strong semantic transfer (79.10% ImageNet accuracy). Our results highlight STELLAR as a versatile sparse representation that bridges discriminative and generative vision by separating semantic identity from spatial geometry. Code is available at <https://github.com/microsoft/STELLAR>.

1. Introduction

Learning visual representations has been a central pursuit in computer vision since the advent of deep learning (Bengio et al., 2013). Modern vision models encode raw pixels into latent features powering nearly all downstream applications. Despite advances from early convolutional networks (Lecun et al., 1998) to ResNets (He et al., 2016) and vision transformers (ViTs) (Dosovitskiy et al., 2020), the geometric format of visual representation has remained largely

unchanged: a dense 2D grid of high-dimensional features, where each vector is tied to a local patch. This design is intuitive, as it mirrors the grid-like arrangement of pixels.

On the other hand, the field faces a longstanding dilemma: the pursuit of a unified, *holistic* representation that excels at both high-level semantic understanding and low-level reconstruction. While this synthesis has been remarkably successful in natural language processing, where predictive objectives such as bidirectional masking (BERT (Devlin et al., 2019)) or autoregressive modeling (GPT (Brown et al., 2020)) also yield strong semantic representations, it does not directly transfer to the vision domain. Representations learned primarily through image reconstruction (e.g., MAE (He et al., 2022), SimMIM (Xie et al., 2022)) often yield semantics that trail behind contemporary state-of-the-art methods. Consequently, recent self-supervised learning (SSL) approaches have largely emphasized one of two directions: those prioritizing pixel-level grounding via reconstruction, and those prioritizing rich semantics via joint-embedding invariance (Van Assel et al., 2025).

We argue that this divergence stems from an *Invariance–Equivariance Tension* in conventional visual SSL representations. Faithful reconstruction requires representations to preserve spatially precise information that changes predictably under transformations such as cropping or shifting. In contrast, high-level semantic recognition benefits from invariance to these transformations. Existing joint-embedding methods such as DINO (Caron et al., 2021) encourage such invariance primarily through global image representations. Although their dense patch features can still retain useful spatial structure and transfer well to dense prediction tasks, the representation most optimized for global semantic alignment is not necessarily the one best suited for spatially faithful reconstruction.

In this work, we show that this tension is not an inevitable trade-off, but can be alleviated by moving beyond a purely dense-grid representation toward a *sparse, factorized latent representation*. This representation family provides a unified latent space in which high-fidelity reconstruction and rich semantics can coexist. Our key insight is that the information necessary to describe a scene can be disentangled into two complementary factors:

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1. **The “What”:** A compact set of sparse latent tokens representing invariant visual concepts.
2. **The “Where”:** A set of equivariant coefficients representing their spatial distributions.

By disentangling these factors through a low-rank matrix factorization form, we enable a “semantic triage”: the model is forced to reconstruct the entire image using a highly compressed bottleneck. This encourages the model to ignore stochastically redundant background pixels and focus on semantically-rich object regions. We propose **STELLAR**, a framework that achieves high-quality reconstruction from as few as 16 tokens while encoding fine-grained semantics in a fully self-supervised manner.

Our contributions are summarized as follows:

- **Sparse Representation:** We propose **STELLAR**, a framework that represents image semantics with as few as 16 concept tokens while retaining spatial information through lightweight localization coefficients, enabling both high-quality reconstruction and fine-grained semantic understanding.
- **SSL Method:** We introduce a training scheme to learn these representations without annotation. By aligning visual concepts across views using optimal transport, we enforce invariance in the “what” factor while adapting the “where” factor, inducing rich semantics.
- **Empirical Observations:** (i) **STELLAR** achieves the strongest joint semantic-reconstruction performance among the evaluated reconstruction-feasible representations, reaching 79.10% IN-1K linear accuracy and 2.60 FID with only 16 semantic tokens. (ii) Sparse factorized training induces fine-grained, region-aware dense features even without explicit dense supervision, yielding strong transfer on downstream dense tasks.

2. Related Work

Self-supervised Learning. Modern SSL is often organized around two dominant paradigms. *Joint Embedding* (JE) methods, such as MoCo (He et al., 2020) and the DINO family (Caron et al., 2021; Oquab et al., 2023), prioritize global invariance via multi-view alignment, yielding strong semantics. Conversely, *Masked Image Modeling* (MIM), exemplified by MAE (He et al., 2022) and SimMIM (Xie et al., 2022), emphasizes spatial equivariance through pixel reconstruction. Hybrid methods such as iBOT (Zhou et al., 2021), DINOv2 (Oquab et al., 2023), CMAE (Huang et al., 2023), and CAN (Mishra et al., 2022) combine reconstruction-style or patch-level objectives with contrastive/joint-embedding learning, while SODA (Hudson et al., 2024) explores representations for supporting both semantic and generative

behavior. **STELLAR** differs by explicitly factorizing the latent feature map into semantic concept tokens and spatial localization, allowing invariance and equivariance to be modeled by separate factors.

Sparse Representation. A growing body of work replaces dense feature maps with compact embeddings. Sparse R-CNN (Sun et al., 2021) and Mask2Former (Cheng et al., 2022) utilize sparse queries for supervised tasks, while BLIP-2 (Li et al., 2023) and TiTok (Yu et al., 2024) employ sparse tokens for vision–language or generative efficiency. SemMAE (Li et al., 2022) utilizes sparse tokens to guide masking using a pretrained teacher. Related object-centric and slot-based models, such as Slot Attention (Locatello et al., 2020), also decompose scenes into a small set of latent slots with spatial assignments, but are typically designed for explicit object discovery or scene decomposition rather than general-purpose SSL transfer. In comparison, **STELLAR** learns sparse tokens as an **explicit factorized latent representation** in a self-supervised manner, while preserving both semantic and reconstructive utility.

Disentanglement & Low-rank Factorization. The assumption that high-dimensional data lie on low-dimensional manifolds is foundational to dictionary learning (Mairal et al., 2008). In deep learning, low-rank constraints are typically applied to weights for efficiency (e.g., LoRA (Hu et al., 2022)). **STELLAR** differs by applying *low-rank factorization to the feature map itself*, disentangling “what” (semantic latents S) from “where” (spatial assignments L).

The Empirical Dilemma. Current vision frameworks face a persistent gap: models excelling at pixel-level reconstruction often produce weaker semantic representations (Zhang et al., 2022; Chen et al., 2024), while those achieving top-tier semantics often reduce or abandon pixel-level reconstruction to avoid low-level shortcuts (Assran et al., 2023; Darcet et al., 2025). We demonstrate that by factorizing the latent representation, it is possible to achieve strong performance on both image understanding and reconstruction.

3. Preliminaries

Representation learning involves encoding an image $X \in \mathcal{X}$ to latent features $Z(X)$ for downstream tasks. Traditionally, vision representations take a *dense* spatial form:

$$Z \in \mathbb{R}^{n \times d},$$

where $n = h \times w$ denotes the number of patches on a dense grid that partitions the image. Each grid location is represented by a feature vector $\mathbf{z}_i := Z_{i,:} \in \mathbb{R}^d$ for $1 \leq i \leq n$. Most vision architectures also incorporate a global representation $\mathbf{z}_0 \in \mathbb{R}^d$, typically obtained via global pooling or a specialized [CLS] token that undergoes self-attention with patch tokens.

Ideally, we want $\mathbf{Z}$ to serve as a *holistic* representation of the image X , which retains sufficient information about the image details, while at the same time possesses rich semantics for downstream tasks. Mathematically, we define such representation as follows:

- **Reconstruction:** There exists a decoder $\mathcal{D}$ such that $\mathcal{D}(\mathbf{Z}(X)) \approx X$. This ensures the representation is spatially and texturally grounded in the physical input.
- **Semantics:** For a downstream task with joint distribution $(X, Y) \sim \mathcal{X} \times \mathcal{Y}$, there exists a simple predictor $f \in \mathcal{F}$ (e.g., a linear layer) such that the expected task loss $\mathbb{E}_{(X,Y)}[\mathcal{L}(f(\mathbf{Z}(X)), Y)]$ is minimized using frozen features. Typically Y reflects human perception.

Current SSL paradigms face an *Invariance-Equivariance Tension* between semantic invariance and spatial equivariance. To learn high-level semantics, Joint Embedding (JE) methods (e.g., the DINO family) encourage invariance across augmented views, including aggressive crops that may retain only a fraction of the original image. On the other hand, reconstruction requires spatial detail, because every pixel shift requires a different set of features for precise reconstruction. This results in representations which are highly *equivariant* to the transformation, i.e. the feature map transforms along with the image transformation.

Let $\mathcal{T}$ be a group of spatial transformations (e.g., translations), and let $t_\theta \in \mathcal{T}$ be parametrized by θ . A representation $\mathbf{Z}(X)$ faces this tension when it is encouraged to satisfy two competing desiderata:

- **Semantic Invariance:** The global readout of the representation should be insensitive to $t_\theta \in \mathcal{T}$:

$$\left\| \frac{\partial}{\partial \theta} \phi(\mathbf{Z}(t_\theta \circ X)) \right\|_F \approx 0.$$

- **Spatial Equivariance:** To allow for high-fidelity reconstruction, the representation must track spatial shifts: $\mathcal{D}(\mathbf{Z}(t_\theta \circ X)) \approx t_\theta \circ X$. Informally, assuming the decoder is locally smooth around $\mathbf{Z}(t_\theta \circ X)$, the chain rule and matrix norm inequalities suggest

$$\left\| \frac{\partial}{\partial \theta} \mathbf{Z}(t_\theta \circ X) \right\|_F \gtrsim \frac{\left\| \frac{\partial(t_\theta \circ X)}{\partial \theta} \right\|_F}{L_D} > 0,$$

where L_D denotes a local Lipschitz constant of the decoder around $\mathbf{Z}(t_\theta \circ X)$.

4. The STELLAR Framework

4.1. Sparse Image Modeling

We now consider an alternative to dense grid-based representations, which describe what appears at each individual

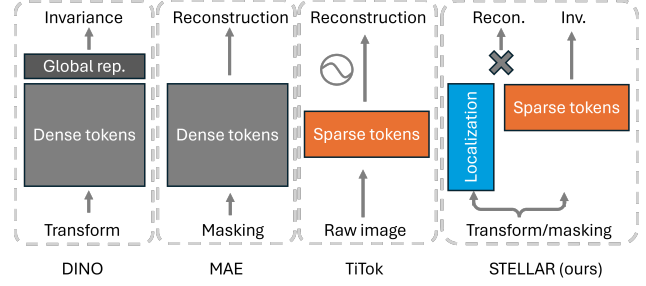


Figure 1. Comparison of latent representation learning paradigms. STELLAR learns a sparse factorized latent that separates semantic concepts from spatial localization, enabling semantic alignment and faithful reconstruction within a single representation.

location. We start from the principle that an image depicts the physical world, which can be understood as a collection of objects located in space.

To begin with, we model an image using a compact set of semantic concept embeddings together with their spatial distributions. Let there be r concept embeddings $s_1, \dots, s_r \in \mathbb{R}^d$, where each s_j is intended to encode a visual concept. The spatial distribution of these concepts is expressed through weights $l_1, \dots, l_n \in \mathbb{R}^r$, where n is the total number of patches.

By constraining $0 \leq l_i \leq 1$ and $\mathbf{1}^\top l_i = 1$, each patch is represented as a convex combination of the concept embeddings: $v_i = \sum_{j=1}^r l_{i,j} s_j$. Thus, the set $\{s_j\}_{j=1}^r$ acts as a compact concept dictionary for constructing local features. In matrix form, the latent representation now takes the form

$$\mathbf{Z}(X) = \mathbf{L}(X)\mathbf{S}(X), \quad (1)$$

where $\mathbf{S} = [s_1, \dots, s_r]^\top \in \mathbb{R}^{r \times d}$ is the *semantic matrix*, and $\mathbf{L} = [l_1, \dots, l_n]^\top \in \mathbb{R}^{n \times r}$ is the *localization matrix*, with the constraint $0 \leq \mathbf{L} \leq 1, \mathbf{L}\mathbf{1}_r = \mathbf{1}_n$. We illustrate the factorized representation in Fig. 1, in contrast to other learning paradigms.

Compared with a canonical dense representation of shape $n \times d$, $\mathbf{Z} = \mathbf{L}\mathbf{S}$ defines a rank-at-most- r factorization of the latent feature map. While the form resembles the low-rank structure used in convex semi-nonnegative matrix factorization (Ding et al., 2008), $\mathbf{S}$ and $\mathbf{L}$ are not obtained from any matrix factorization algorithm, but are instead direct outputs of the encoder forward pass, allowing end-to-end training using SSL objectives.

4.2. Equivariant Partitioning

The factorized form in equation 1 provides a compact latent interface (storing $(\mathbf{L}, \mathbf{S})$ requires $r(n + d)$ values instead of nd for a dense representation. With $r = 16$ tokens for a ViT-Base model at 224×224 resolution, this yields about a 90% reduction in representation size). More importantly,

this factorization provides a mechanism to alleviate the invariance–equivariance tension. The spatial transformation is now partitioned as follows:

$$\underbrace{\frac{\partial \mathbf{Z}(t_\theta \circ X)}{\partial \theta}}_{\text{Total Equivariance}} = \underbrace{\left(\frac{\partial \mathbf{L}(t_\theta \circ X)}{\partial \theta} \right) \mathbf{S}}_{\text{Spatial Equivariance}} + \underbrace{\mathbf{L} \left(\frac{\partial \mathbf{S}(t_\theta \circ X)}{\partial \theta} \right)}_{\text{Semantic Variance} \approx 0}. \quad (2)$$

With spatial and semantic information factorized, spatial equivariance can be primarily absorbed by the localization matrix $\mathbf{L}$, while the semantic matrix $\mathbf{S}$ can remain comparatively stable across transformations. We require that $\mathbf{Z}(X)$ can reconstruct the image by minimizing

$$\mathcal{L}_{recon} = \ell(\mathcal{D}(\mathbf{L}(X)\mathbf{S}(X)), X). \quad (3)$$

This low-rank reconstruction constrains each local feature to be composed from only r concept tokens, forcing the model to explain the image through a compact semantic dictionary while retaining spatial information in $\mathbf{L}$.

4.3. Vision Concept Clustering

To encourage sparse tokens to represent transferable vision concepts, we structure them into K learnable prototypes $\mathbf{c}_1, \dots, \mathbf{c}_K \in \mathbb{R}^p$. As shown in Fig. 2, a backbone encoder $\mathcal{E}$ maps a mini-batch of m images into sparse features $\mathbf{S}^1, \dots, \mathbf{S}^m$. Each token is projected onto the unit sphere $\mathbb{S}^{p-1}$ via a normalized projector $h : \mathbb{R}^d \rightarrow \mathbb{S}^{p-1}$, and its similarity to prototypes $\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_K]$ gives logits

$$\lambda_j^i = [\mathbf{c}_1 \cdot h(\mathbf{s}_j^i), \dots, \mathbf{c}_K \cdot h(\mathbf{s}_j^i)], \quad j = 1, \dots, r. \quad (4)$$

Soft assignments over the prototypes are obtained with

$$q_{j,k}^i = \frac{\exp(\lambda_{j,k}^i / \tau)}{\sum_{k'=1}^K \exp(\lambda_{j,k'}^i / \tau)}, \quad (5)$$

where τ controls the sharpness. Direct entropy minimization of $q_{j,k}^i$ is unstable due to non-convexity and empty clusters. Following (Caron et al., 2020; Darcet et al., 2025), we compute balanced assignments $\tilde{q}_{j,k}^i$ from $q_{j,k}^i$ using the Sinkhorn-Knopp algorithm with stop-gradient, and minimize

$$\mathcal{L}_{cluster} = -\frac{1}{mr} \sum_{i=1}^m \sum_{j=1}^r \sum_{k=1}^K \tilde{q}_{j,k}^i \log q_{j,k}^i. \quad (6)$$

The resulting balanced assignments are treated as stop-gradient targets for the clustering objective.

4.4. Set Concept Alignment

To achieve the semantic invariance in equation 2, we align the sparse tokens $\mathbf{s}_1, \dots, \mathbf{s}_r$ obtained from a transformed

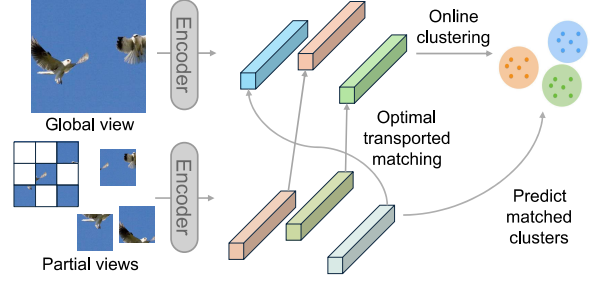


Figure 2. Concept clustering and set-alignment workflow. Sparse tokens are assigned to balanced visual prototypes, and tokens from transformed views are matched to those from the global view using optimal transport. The resulting assignments provide view-consistent supervision for learning invariant semantic concepts.

view (e.g. masking or cropping) to the ones from the global view $\mathbf{s}_1, \dots, \mathbf{s}_r$. However, this set concept alignment problem is more challenging than global representation alignment in traditional JE methods because there is no inherent ordering among the r tokens. To solve the problem, we apply optimal transport with the cost matrix

$$\Theta_{j'j} = \|\mathbf{s}_{j'}' - \mathbf{s}_j\|_2. \quad (7)$$

We solve for an assignment matrix $\mathbf{P}$ via entropy-regularized optimal transport:

$$\min_{\mathbf{P} \geq 0} \sum_{j',j} \mathbf{P}_{j'j} \Theta_{j'j} - \epsilon H(\mathbf{P}), \quad (8)$$

$$\text{s.t. } \mathbf{P} \mathbf{1}_r = \mathbf{P}^T \mathbf{1}_r = \frac{1}{r} \mathbf{1}_r, \quad (9)$$

with $H(\mathbf{P}) = -\sum_{j',j} \mathbf{P}_{j'j} \log \mathbf{P}_{j'j}$. We solve for $\mathbf{P}$ using the Sinkhorn algorithm, and define the matching $\sigma(j') := \arg\max_j \mathbf{P}_{j'j}$. Compared to bipartite matching algorithms such as Hungarian matching, which are widely used in prior work, this Sinkhorn-based procedure is up to 100× faster in our implementation, as analyzed in the Appendix B.8.

We then compute prototype assignments for the transformed view tokens $q_{j'}' = \text{softmax}(\mathbf{C}^T h(\mathbf{s}_{j'}') / \tau)$, and minimize the set concept alignment loss

$$\mathcal{L}_{align} = -\frac{1}{r} \sum_{j'=1}^r \sum_{k=1}^K \tilde{q}_{\sigma(j'),k} \log q_{j',k}' . \quad (10)$$

Optionally, we use the same framework to cluster and align the CLS token with its own projector and prototypes, similar to previous JE methods. However, we do not use it for reconstruction. We also apply KoLeo regularization (Sablajrolles et al., 2018) on the normalized sparse tokens $\bar{\mathbf{s}}_j := \mathbf{s}_j / \|\mathbf{s}_j\|$ from the same image to encourage concept diversification:

$$\mathcal{L}_{KoLeo} = -\frac{1}{r} \sum_{j=1}^r \log \left(\min_{j' \neq j} \frac{1}{2} \|\bar{\mathbf{s}}_j - \bar{\mathbf{s}}_{j'}\|_2 \right). \quad (11)$$

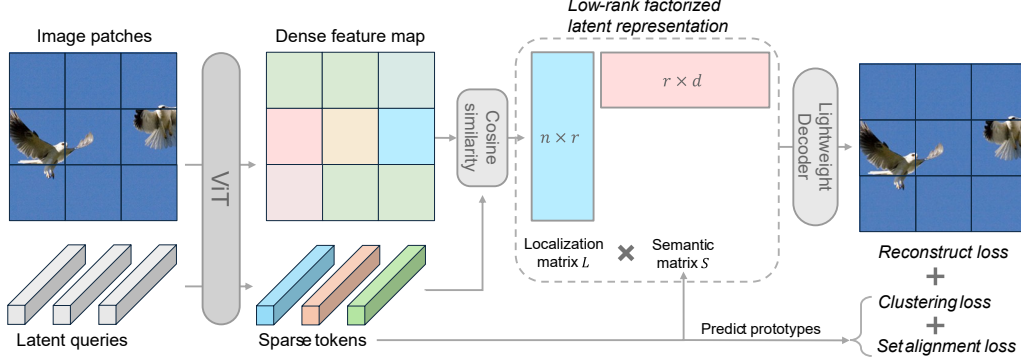


Figure 3. Overview of the STELLAR framework. A ViT encoder produces dense patch features and a small set of sparse concept tokens. A localization matrix assigns each patch to these concepts, yielding a low-rank representation $Z = LS$ for reconstruction. Self-supervised clustering and set-alignment losses are applied to the sparse tokens, encouraging them to form transferable visual concepts.

All together, we jointly optimize the following objectives by training the encoder $\mathcal{E}$, decoder $\mathcal{D}$, projector h , and prototypes $\mathcal{C}$ jointly with the final objective:

$$\min_{\mathcal{E}, \mathcal{D}, h, \mathcal{C}} a_1 \mathcal{L}_{\text{recon}} + a_2 \mathcal{L}_{\text{cluster}} + a_3 \mathcal{L}_{\text{align}} + a_4 \mathcal{L}_{\text{cluster-cls}} + a_5 \mathcal{L}_{\text{align-cls}} + a_6 \mathcal{L}_{\text{KoLeo}}. \quad (12)$$

In summary, we propose a sparse vision representation $(S, L) = \mathcal{E}(X)$ that explicitly disentangles semantic concepts from their spatial distributions, enabling the latent variables to support both pixel-level reconstruction and high-level semantic understanding. We introduce a simple encoder design to obtain these latent variables and SSL objectives to shape them into transferable visual concepts. The full framework overview is illustrated in Fig. 3.

We refer to our framework of learning the spatial-semantic factorized representation $Z(X) = L(X)S(X)$ as **S**parse **T**oken **E**xtraction and **L**ocalization with **L**ow-rank **A**pproximated **R**econstruction (STELLAR).

4.5. Model Design

We note that the framework only specifies the latent space, and does not prescribe any specific encoder or decoder architecture. In this work, we adopt a simple design with common modules and model architectures to obtain S and L as described below. We deliberately use standard components to focus on the representation form and SSL objectives, rather than on architectural novelty.

For the encoder part, we use a ViT (Dosovitskiy et al., 2020) as the backbone, and equip it with r learnable latent query vectors, which are passed to the transformer blocks alongside the patch tokens. Processed by the ViT jointly, the latent queries produce sparse tokens $S \in \mathbb{R}^{r \times d}$.

To obtain the localization matrix $L \in \mathbb{R}^{n \times r}$ associated with

the sparse tokens, we use the dense feature map $U \in \mathbb{R}^{n \times d}$ output from the image patches. We project both S and U into a shared embedding space and compute their pairwise cosine similarities, followed by a softmax normalization with temperature τ_{spatial} along the sparse-token dimension:

$$L = \text{softmax}(\text{cossim}(UW_1, SW_2)/\tau_{\text{spatial}}). \quad (13)$$

W_1 and W_2 are learnable linear projections, and τ_{spatial} controls the sharpness of the spatial distribution. We note that this mapping is structurally similar to the attention weights obtained in a single-head cross-attention layer, up to the use of L2 normalization and an explicit temperature parameter. Therefore, the latent representation $Z = LS$ can be viewed as rebuilding a dense feature map for reconstruction by cross-attending to only r sparse concept tokens.

All together, the encoder $\mathcal{E}$ includes ViT transformer blocks, r learnable latent query vectors, and projection layers W_1, W_2 . The decoder $\mathcal{D}$ is a lightweight 6-layer ViT that reconstructs image patches.

The STELLAR framework can be initialized from a pre-trained ViT, such as MAE or DINO, to reshape a strong vision prior into a sparse holistic representation. It can also be trained from random initialization and achieve competitive spatial, semantic, and reconstruction quality. We provide a detailed analysis in the ablation study.

The resulting model exposes several complementary representation readouts. We use the sparse concept tokens S as the compact semantic representation, while L provides their spatial support and LS forms the factorized representation used for reconstruction. The dense backbone feature map U is an intermediate encoder representation and is retained when full-resolution dense features are required.

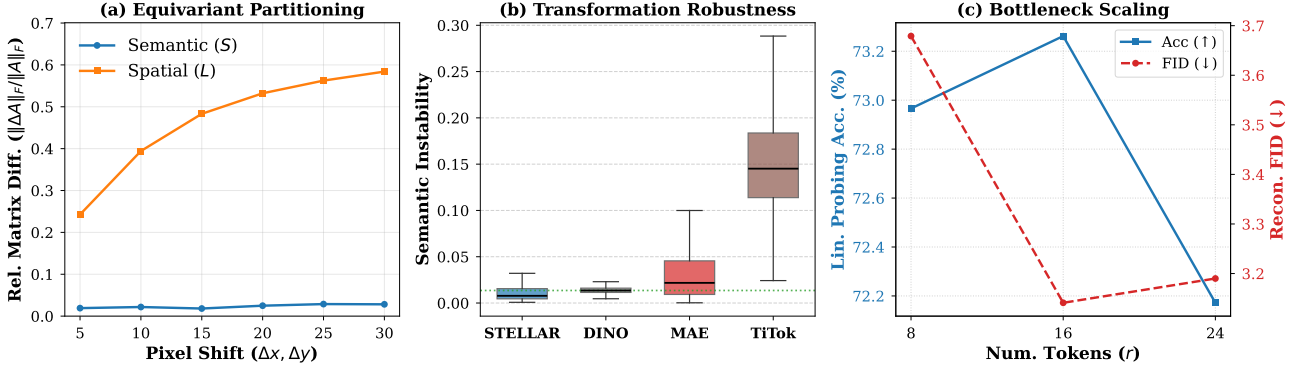


Figure 4. Analysis of the STELLAR representation. (a) Relative matrix difference in L and S under controlled pixel shifts in the input image. (b) Cosine distance of latent representations under random crops retaining 50-100% of the image. (c) Impact of the number of sparse tokens r on reconstruction and semantic quality.

5. Experiments

We train STELLAR on ImageNet-1K (Deng et al., 2009) without labels. The encoder is a vanilla ViT (Dosovitskiy et al., 2020) augmented with 8–24 learnable latent queries that produce sparse tokens. A lightweight 6-layer ViT serves as the decoder, predicting MaskGIT-VQGAN tokens (Esser et al., 2021; Chang et al., 2022). When using a pretrained backbone, we ensure that the pretraining was also performed only on ImageNet-1K. We use MAE as the default prior and study the effect of different initialization modes in the ablation study. When training from random initialization, we use a momentum-updated encoder to encode the target assignments $\tilde{q}$ in equation 6 and equation 10, following Grill et al. (2020); Caron et al. (2021).

5.1. Probing the Factorized Representation

We designed a series of experiments to analyze the factorized representation $Z = LS$ learned by STELLAR.

Experiment 1: Equivariant Partitioning We built a controllable and parametrized spatial transformation group to examine the equivariant partitioning in equation 2. Given an image, we take a crop and shift it gradually from 5 to 30 pixels, either horizontally or vertically. We calculated the relative matrix difference $\frac{\|S(t_\theta \circ X) - S(X)\|_F}{\|S(X)\|_F}$ and $\frac{\|L(t_\theta \circ X) - L(X)\|_F}{\|L(X)\|_F}$. Optimal transport in equation 8 is used to match the token ordering. As shown in Fig. 4(a), the semantic matrix S stays nearly invariant, while the spatial localization matrix L changes continuously with the spatial shift, supporting the effectiveness of equivariant partitioning in the factorized representation.

Experiment 2: Transformation Robustness Next we compare the semantic stability of STELLAR representation with baseline models under random resized cropping

at scale 50-100%. We calculate the cosine distance from the feature of the untransformed image as a measure of semantic instability. For DINO and MAE, we used mean-pooled dense features, and for sparse models STELLAR and TiTok, we used mean-pooled sparse tokens. As shown in Fig. 4(b), the sparse tokens of STELLAR exhibit transformation robustness comparable to DINO. As expected, reconstruction-based models such as MAE and TiTok show higher sensitivity to spatial transformations. In particular, the non-factorized sparse representation from TiTok is highly unstable, where semantic and spatial information being encoded in the same latent space for reconstruction.

Experiment 3: Effect of Low-rank Bottleneck The number of sparse tokens r controls the rank bottleneck of the latent representation, with $\text{rank}(Z) \leq r$. We varied r from 8 to 24 and evaluated reconstruction with FID (Heusel et al., 2017) and semantics with linear probe on mean-pooled sparse tokens. As shown in Fig. 4(c), the accuracy decreases as r increases, while reconstruction improves, showing a trade-off as the rank bottleneck is varied. Empirically, $r = 16$ provides a good operating point, achieving both strong semantics and high-quality reconstruction, and we use it as the default for the remaining experiments.

Finally, we visualize the factorized representation in Fig. 5. We show the spatial localization (thresholded by $1/r$) of a sparse token in the image, and the top retrieved semantic concepts from the training dataset.

5.2. Evaluating Holistic Representation

Next, we examine the semantic quality and reconstruction potential of representations from different models. For MAE, DINO, and STELLAR, we use a decoder-controlled protocol: the encoder is frozen, and the same lightweight STELLAR decoder is trained on top of the frozen features to reconstruct the image. For STELLAR, we also fine-tune

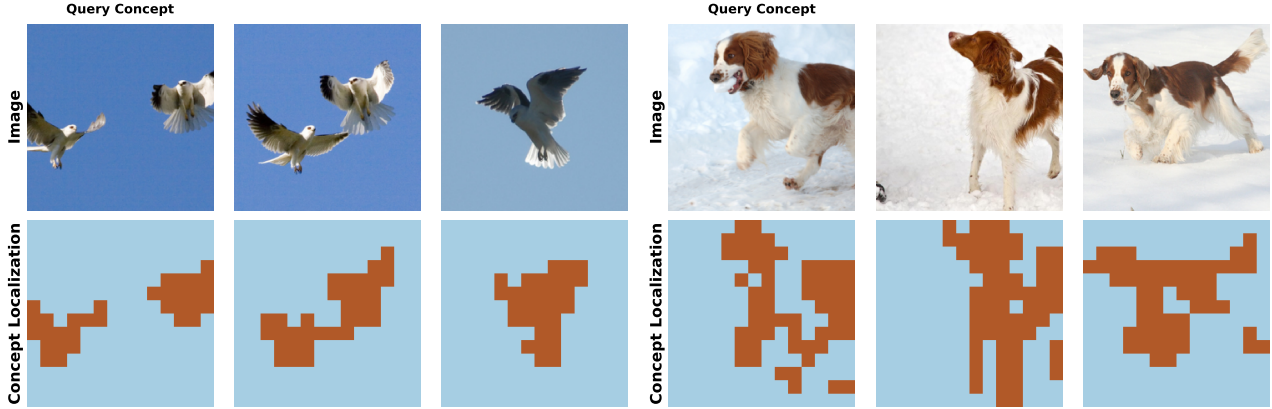


Figure 5. Vision concept retrieval with sparse concept tokens. We show both the image and the spatial localization of each concept.

its decoder with the encoder frozen under the same protocol. TiTok is the exception: because its sparse latent is not spatially factorized, we report results using its native decoder, which is a full-sized ViT compared to the 6-layer decoder used in STELLAR.

As shown in Table 1, STELLAR achieves the strongest joint trade-off between semantics and reconstruction among reconstruction-feasible representations. The linear probing and k-NN accuracy of STELLAR surpass all reconstruction-feasible baselines, while trailing the global CLS token from DINO, which is not itself a reconstruction-feasible latent.

Table 1. Reconstruction and semantic metrics on IN1K of STELLAR (ours) and baseline models. For reference, we also report semantic metrics of the global representation from DINO and a larger STELLAR model. Best main results are shown in **bold**. Model sizes are ViT-B by default, with larger sizes indicated in parentheses. *: TiTok uses its native larger ViT decoder.

MODEL	# TKS	RECONSTRUCTION		SEMANTICS	
		FID ↓	LPIPS ↓	LIN.	KNN
DINO	1	-	-	76.46	74.69
DINO	196	3.27	0.2121	70.31	54.41
MAE	196	3.02	0.2071	66.32	25.82
TiTok*	32	2.75	0.3281	33.42	7.30
TiTok*	64	1.99	0.2571	32.87	7.29
OURS	16	3.06	0.2077	73.26	67.25
OURS	196	2.85	0.2085	72.21	64.71
OURS(H)	16	2.60	0.1729	79.10	77.31

On reconstruction, STELLAR achieves FID and LPIPS (Zhang et al., 2018) comparable to the dense feature map from MAE, while reducing the latent size by about 90%. Although TiTok achieves lower FID with a much larger decoder, it has substantially worse LPIPS, indicating weaker per-image spatial consistency. In contrast, STELLAR exhibits stronger reconstruction locality with fewer tokens. The full-rank dense feature map

U from STELLAR achieves lower reconstruction FID, but with slightly reduced semantic quality. Finally, scaling STELLAR to ViT-H further improves both reconstruction and semantic quality.

5.3. Benchmarking Image Understanding

Lastly, we benchmark STELLAR in classical image understanding tasks with linear probing on frozen features, comparing against other ImageNet-pretrained SSL models. We report results for classification on ImageNet-1K (IN1K), Oxford-IIIT Pet (Pets) (Parkhi et al., 2012), Food-101 (Food) (Bossard et al., 2014), and GlaS (Sirinukunwattana et al., 2016) for cancer grade classification in histopathology. Segmentation benchmarks include ADE20K (Zhou et al., 2017b), Cityscapes (Cordts et al., 2016), and Pascal VOC (Everingham et al., 2010). For broader context, we also include AIM (El-Nouby et al., 2024) and DINOv2 (Oquab et al., 2023), which leverage substantially larger training corpora (100–1000× more images).

As shown in Table 2, the feature map from STELLAR achieves superior performance on ADE20K and Pascal VOC, showing strong fine-grained understanding despite not applying SSL objectives directly to the dense feature map U . The results show that sparse factorized modeling implicitly organizes the feature map into semantic regions. Because the image must be reconstructed from a small set of sparse tokens and their spatial localization, each token is encouraged to capture coherent visual content that can be reused across different spatial parts of the scene. This leads to region-aware dense representations. While MAE leads on Cityscapes, STELLAR follows closely with performance comparable to MAE and DINOv2.

On global image understanding tasks, STELLAR achieves the highest IN1K accuracy at large model scale, but smaller variants underperform methods such as DINO that explicitly optimize global representations. Overall, STELLAR outper-

Table 2. Evaluation of Fine-grained and Global Image Understanding. We evaluate semantic segmentation (mIoU %) and classification accuracy (%) via linear probing on frozen features. For segmentation, we use the dense feature map from the backbone for all models. All numbers are obtained from our own evaluation using official released checkpoints under a consistent protocol. **Bold**: best among ImageNet-trained models. Underline: best within the corresponding architectural class/scale group.

Model	Arch.	SSL Method		Segmentation (mIoU)			Classification (Acc)			
		Target	Objective	ADE20K	CitySc	VOC	IN1K	Pets	Food	GlaS
Semantic-Centric (Joint Embedding / Invariance)										
BYOL	RN-50	GLOBAL	JE	18.43	<u>18.66</u>	63.89	<u>70.39</u>	<u>82.77</u>	<u>64.57</u>	<u>95.00</u>
MoCo v3	ViT-B	GLOBAL	JE	29.45	25.13	74.08	74.31	91.14	77.47	97.50
DINO	ViT-B	GLOBAL	JE	26.87	26.82	79.29	<u>76.46</u>	93.84	<u>79.28</u>	95.00
MSN	ViT-B	GLOBAL	JE	26.66	25.39	68.59	73.65	75.91	68.93	92.50
DENSECL	RN-50	DENSE	JE	<u>23.08</u>	18.63	<u>70.95</u>	61.10	72.99	59.16	85.00
DATA2VEC	ViT-B	DENSE	LATENT MIM	22.03	23.49	61.33	54.90	26.47	34.40	73.75
SIAMESEIM	ViT-B	DENSE	LATENT MIM	29.24	26.52	81.38	74.97	91.61	71.01	91.25
I-JEPA	ViT-H	DENSE	LATENT MIM	21.57	18.59	74.13	71.72	84.68	70.34	87.50
iBOT	ViT-B	GL+DE	JE+MIM	<u>31.78</u>	25.69	77.06	76.40	92.40	78.08	96.25
iBOT	ViT-L	GL+DE	JE+MIM	33.26	26.37	77.57	<u>78.53</u>	92.12	81.07	96.25
Image-Centric (Reconstruction)										
BEiT	ViT-B	DENSE	TOKEN MIM	11.58	18.90	27.44	32.94	36.20	54.49	90.00
BEiT	ViT-L	DENSE	TOKEN MIM	12.64	20.37	25.48	36.77	36.71	56.03	90.00
SIMMIM	SWIN-B	DENSE	PIXEL MIM	12.46	17.23	35.14	24.77	27.39	40.94	77.50
MAE	ViT-B	DENSE	PIXEL MIM	30.91	<u>29.44</u>	76.43	66.32	81.58	70.40	93.75
MAE	ViT-L	DENSE	PIXEL MIM	<u>34.36</u>	<u>32.53</u>	77.79	73.09	84.30	76.22	95.00
MAE	ViT-H	DENSE	PIXEL MIM	36.16	35.21	78.07	75.22	84.96	<u>78.36</u>	<u>95.00</u>
SEMMAE	ViT-B	DENSE	PIXEL MIM	3.52	25.48	48.33	43.84	56.99	58.90	92.50
TiTOK-64	ViT-B	SPARSE	SPARSE REC	–	–	–	32.87	42.06	43.68	97.50
TiTOK-32	ViT-L	SPARSE	SPARSE REC	–	–	–	33.42	27.83	38.83	78.75
Our Method (Sparse Factorized Modeling)										
STELLAR	ViT-B	SPARSE	JE+REC	31.33	27.74	<u>81.83</u>	73.26	89.70	74.09	95.00
STELLAR	ViT-L	SPARSE	JE+REC	34.02	31.32	85.90	76.94	<u>92.53</u>	74.78	97.50
STELLAR	ViT-H	SPARSE	JE+REC	36.66	33.30	<u>85.66</u>	79.10	<u>92.53</u>	77.43	92.50
Larger Scale Pretraining Beyond ImageNet (Reference Only)										
AIM	600 M	DENSE	IMAGE AR	29.00	27.04	64.55	63.78	64.68	75.19	98.75
AIM	1 B	DENSE	IMAGE AR	29.59	27.05	63.90	66.86	64.21	77.96	96.25
DINOv2	ViT-B*	GL+DE	JE+MIM	40.10	34.66	89.52	82.82	95.59	91.08	98.75
DINOv2	ViT-L*	GL+DE	JE+MIM	40.45	32.07	89.19	84.23	96.08	92.94	98.75

forms reconstruction-based models and many JE methods, while still trailing the strongest JE models on pure global semantics. Since STELLAR does not model the image as a single global concept, averaging sparse tokens can dilute discriminative information, which is particularly detrimental on object-centric datasets such as Pets and Food. Interestingly, on histopathology images involving complex tissue microenvironments, STELLAR achieves top performance. These results suggest that STELLAR is especially effective for complex, multi-region scenes, while global classification on simple object-centric datasets remains more challenging.

5.4. Ablation Analysis

Low-rank reconstruction. As shown in Table 3, removing the low-rank reconstruction objective (A) reduces both global and fine-grained understanding. Since the remaining objectives resemble typical SSL methods, the model still retains reasonable global performance, but fine-grained understanding suffers more. This indicates that low-rank

reconstruction encourages sparse tokens to serve as holistic representations covering the entire image.

Concept clustering. Eliminating online clustering and set alignment (B) leads to a sharp drop in understanding, highlighting the necessity of structuring sparse tokens into view-invariant concepts. Even when the alignment loss is present (D), removing the clustering loss still leads to degradation.

Set alignment. The training collapsed when training with only reconstruction and clustering (C), underscoring the critical role of set concept alignment. Additional alignment on the CLS token (E) primarily benefits global classification but has limited effect on spatial grounding. Finally, KoLeo regularization (F) consistently improves all tasks at a similar level. Interestingly, the absence of either concept clustering or set alignment leads to a sharp drop in performance.

Overall, the sharp drops without concept clustering or set alignment show that the factorized form alone is not sufficient: these objectives are necessary to turn sparse tokens

Table 3. Ablation. We isolate the impact of each objective on semantic abstraction (IN1K) and spatial grounding (ADE20K), and reconstruction (FID). *Default* denotes the full STELLAR framework. All results are based on ViT-B.

	Recon.	Cluster	Set Align	CLS Align	KoLeo	FID ↓	IN1K ↑	ADE ↑
DEFAULT	✓	✓	✓	✓	✓	3.14	73.26	31.33
<i>Impact of Individual Components</i>								
(A)	✗	✓	✓	✓	✓	—	72.44 (-0.82)	29.94 (-1.39)
(B)	✓	✗	✗	✗	✓	3.21 (+0.07)	52.07 (-21.19)	20.46 (-10.87)
(C)	✓	✓	✗	✗	✓	8.95 (+5.81)	2.73 (-70.53)	1.93 (-29.39)
(D)	✓	✗	✓	✓	✓	3.62 (+0.48)	42.14 (-31.12)	18.90 (-12.43)
(E)	✓	✓	✓	✗	✓	3.26 (+0.12)	70.79 (-2.47)	30.20 (-1.12)
(F)	✓	✓	✓	✓	✗	3.25 (+0.11)	72.05 (-1.21)	30.10 (-1.23)

into stable visual concepts. Together, the representation form and the SSL objectives enable STELLAR to learn transferable “what” tokens while preserving spatial grounding through “where” coefficients.

Effect of foundation prior. We ablate different initialization modes for STELLAR in Table 4. STELLAR substantially improves the semantic quality of an MAE prior, increasing IN1K linear probing from 66.32 to 73.26, and improves the spatial grounding of a DINO prior, increasing ADE20K from 26.87 to 28.17. When trained from random initialization, STELLAR still reaches MAE-level semantic performance and DINO-level spatial understanding, while maintaining comparable reconstruction quality. These results suggest that the STELLAR objectives can learn a useful factorized representation from scratch, while strong pretrained priors further improve the final representation.

Table 4. Evaluating STELLAR trained from different foundational priors. *Base* represents the performance of the original backbone.

Prior	Recon FID ↓	Semantic (IN1K)		Spatial (ADE20K)	
		BASE	+STELLAR	BASE	+STELLAR
MAE	3.14	66.32	73.26 (+6.9)	30.91	31.33 (+0.4)
DINO	3.31	76.46	73.31 (-3.2)	26.87	28.17 (+1.3)
RAND	3.21	—	65.28	—	28.10

6. Discussion and Concluding Remarks

We have shown that the tension between *semantic abstraction* and *spatial grounding* can be alleviated by changing the form of visual representation. STELLAR factorizes the latent space into sparse “what” and “where” components: semantic tokens S remain comparatively stable across views, while the localization matrix L captures spatial variation needed for reconstruction. This sparse bottleneck induces a form of *semantic triage*, encouraging the model to explain an image through a small set of reusable, spatially grounded concepts. Rather than discarding spatial information, this reorganization allows semantic invariance and spatial equiv-

ariance to be modeled through separate factors.

Beyond self-supervised pretraining, sparse factorized representations may provide a useful interface for multimodal models. Modern vision-language systems often pass hundreds of dense visual tokens to a language model, many of which are spatially redundant. STELLAR suggests an alternative with compact semantic concepts together with their spatial support, potentially improving visual token efficiency and cross-modal grounding while retaining information needed for spatial reasoning.

A particularly promising direction is to extend this representation from images to *video and world modeling*. Across time, semantic identity is often relatively persistent while spatial configuration and appearance evolve, suggesting a natural temporal analogue of the “what-where” factorization. Sparse concept tokens could provide persistent entities or semantic state variables, while their localization coefficients capture spatial dynamics. A predictive model could then learn how these factors evolve over time, potentially modeling long-horizon dynamics in a substantially more compact latent space than dense spatiotemporal tokens. More broadly, learning which aspects of a visual state should remain stable and which should evolve may provide a useful representation principle for world models.

Several limitations remain. STELLAR obtains its strongest results when reshaping a pretrained vision prior, although it can also be trained from scratch. The encoder still computes dense patch features to obtain the localization matrix, so its primary efficiency benefit lies in representation compression and downstream token processing rather than encoder-side compute. Finally, the optimal number of sparse tokens may depend on image complexity, resolution, and downstream use cases. Extending sparse factorized representations to higher-resolution images and open-vocabulary multimodal systems is an important direction for future work.

Overall, our results suggest that sparse factorized tokens are a promising representation format for jointly supporting semantic abstraction, spatial grounding, reconstruction, and efficient multimodal interfacing.

Impact Statement

This work advances self-supervised visual representation learning by introducing compact, semantically grounded, and reconstruction-capable image representations. Potential positive impacts include more efficient visual tokenization, improved representation reuse across recognition and reconstruction tasks, and more interpretable visual features for multimodal systems.

As with other general-purpose visual representation learning methods, STELLAR may also be adapted to sensitive applications. These risks are not specific to our method, but improved semantic understanding and reconstruction capability should be deployed with appropriate privacy, data-governance, and application-specific safeguards.

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A. Implementation Details

A.1. STELLAR Training

We train STELLAR with ViT-Base, ViT-Large, and ViT-Huge backbones, each augmented with learnable latent queries, projection layers, prototype heads, and a lightweight 6-layer ViT decoder. Unless otherwise specified, we initialize the ViT backbone from a public MAE checkpoint pretrained on ImageNet-1K, while all newly introduced STELLAR components are randomly initialized. We train STELLAR-B for 150 epochs, STELLAR-L for 100 epochs, and STELLAR-H for 50 epochs. All models are trained on 16 NVIDIA A100-80GB GPUs with a per-GPU batch size of 128, resulting in a total batch size of 2048. We use AdamW (Loshchilov, 2017) with base learning rate 1.5×10^{-4} for STELLAR-B and 5×10^{-5} for STELLAR-L/H.

For concept clustering, we use $K = 16384$ prototypes for sparse tokens and another $K = 16384$ prototypes for the CLS token. The projector h is a 2-layer MLP before the prototype layer. Balanced prototype assignments are computed with 3 Sinkhorn-Knopp iterations using stop-gradient targets. The spatial localization matrix $\mathbf{L}$ is computed from sparse-dense cosine similarities with softmax temperature $\tau_{\text{spatial}} = 0.06$.

For data augmentation, each image is transformed into multiple views. We use 6–8 randomly masked views for sparse-token set alignment, and an additional 6–8 local crops for CLS-token alignment. Global views are sampled with random-resized-crop scale between 36% and 100%, while local views use scale between 6% and 36%. We also apply standard photometric augmentations, including color jittering, grayscale conversion, and Gaussian blurring.

For the random-initialization ablation, we train STELLAR from scratch and use an exponential moving average (EMA) momentum encoder during the warm-up stage to produce target prototype assignments in equation 6 and equation 10. The EMA encoder includes the ViT backbone, latent queries, projection layers, and prototype heads, and is updated with momentum 0.996. The momentum encoder processes a global view of the image to produce target assignments for both clustering and set alignment. We use masking ratio 0.6 during the EMA warm-up stage and 0.8 during standard training. The random-initialization model is trained for 150 epochs of EMA warm-up followed by 75 epochs of standard training.

For the loss weights in equation 12, we set $a_1 = 0.05$ for the reconstruction loss, which is implemented as cross-entropy over predicted VQGAN codebook indices. We set $a_2 = a_3 = a_4 = a_5 = 1$ for the sparse-token clustering, sparse-token set alignment, CLS-token clustering, and CLS-token alignment losses. The sparse-token set alignment loss is computed only on randomly masked views, while the CLS-token alignment loss is computed on both masked views and local crops, with the two contributions averaged separately for each view type and then summed. We set the KoLeo regularization weight to $a_6 = 0.1$. KoLeo regularization is applied to both global and masked views. The losses are averaged separately for each view type and then summed.

A.2. Evaluation Protocol

For STELLAR and all baseline models, we evaluate frozen features using linear probing. For classification tasks, we apply layer normalization followed by a single linear classifier. For segmentation tasks, we apply batch normalization followed by a linear classifier at each patch location. For every benchmark, we reserve 10% of the training set for validation and tune hyperparameters on this split. We sweep learning rates in $\{1, 2, 5\} \times \{10^{-5}, 10^{-4}, 10^{-3}\}$ and 10^{-2} , and batch sizes in $\{64, 128, 256, 512, 1024, 2048, 4096, 8192\}$.

Since different SSL methods optimize different parts of the representation, we evaluate classification using the feature type on which the corresponding SSL objective is primarily applied. For example, we use the global CLS token for DINO and dense patch tokens for MAE. For STELLAR, we use mean-pooled sparse tokens by default. For segmentation, we use the dense feature map from the backbone for all methods, ensuring a consistent dense-prediction protocol across models.

We further evaluate how linear probing accuracy changes with token choice. As shown in Table 5, the features directly optimized by the SSL objective usually provide the best linear probing performance. DINO relies strongly on the global token, while MAE is slightly better with dense features. iBOT performs best with the global token despite also using masked image modeling. In contrast, STELLAR achieves strong performance from sparse tokens while its dense features remain competitive, suggesting that sparse factorized training also shapes the dense backbone representation.

Table 5. ImageNet-1K linear probing accuracy (%) using different feature readouts. We mark in **bold** the features on which the corresponding SSL objective is primarily applied, as well as the best accuracy for each method.

	DINO		MAE		iBOT		STELLAR (ours)		
Feature	Global	Dense	Global	Dense	Global	Dense	Gl.+De.	Sparse	Dense
Lin. Acc.	76.46	70.31	65.61	66.32	76.40	71.44	71.58	73.26	72.21

Table 6 summarizes the baseline models used in our evaluation and categorizes them by their self-supervised learning objective, the prediction space, and the type of tokens on which the SSL objective is primarily applied. This taxonomy helps clarify the comparison in Table 2: different methods optimize different parts of the representation, e.g., global tokens for joint-embedding methods, dense patch tokens for reconstruction-based methods, and sparse tokens for STELLAR.

Table 6. List of baseline models and SSL method type.

Model	Reference	Method	SSL space	SSL tokens
BYOL	(Grill et al., 2020)	augmentation alignment	latent	global
MoCo v3	(Chen et al., 2021)	contrastive learning	latent	global
DINO	(Caron et al., 2021)	augmentation alignment	latent	global
MSN	(Assran et al., 2022)	masked alignment	latent	global
DenseCL	(Wang et al., 2021)	contrastive learning	latent	dense
Data2Vec	(Baeviski et al., 2022)	latent MIM	latent	dense
SiameseIM	(Tao et al., 2023)	latent MIM	latent	dense
IJEPA	(Assran et al., 2023)	latent MIM	latent	dense
iBOT	(Zhou et al., 2021)	align + latent MIM	latent	global+dense
BEIT	(Bao et al., 2021)	token MIM	image	dense
SimMIM	(Xie et al., 2022)	pixel MIM	image	dense
MAE	(He et al., 2022)	pixel MIM	image	dense
SemMAE	(Li et al., 2022)	pixel MIM	image	dense
TiTok	(Yu et al., 2024)	reconstruction + clustering	image	sparse
AIM	(El-Nouby et al., 2024)	autoregressive	image	dense
DINOv2	(Oquab et al., 2023)	align + latent MIM	latent	global+dense

B. Additional Results

B.1. Additional Reconstruction Metrics

In addition to FID and LPIPS reported in the main text, we further evaluate per-image reconstruction fidelity using PSNR and SSIM. These metrics provide a complementary view of whether the sparse latent preserves image-specific spatial structure, rather than only producing distributionally plausible reconstructions. As shown in Table 7, STELLAR’s low-rank representation achieves PSNR and SSIM comparable to or slightly better than MAE and DINO under the same decoder-controlled protocol. In contrast, although TiTok achieves lower FID with its native larger decoder, it obtains substantially worse LPIPS, PSNR, and SSIM, suggesting weaker per-image fidelity. These results support the conclusion that STELLAR preserves spatial locality well despite using only 16 sparse tokens.

Table 7. **Additional reconstruction metrics on ImageNet-1K.** We report PSNR and SSIM in addition to FID and LPIPS. For DINO, MAE, and STELLAR, reconstruction is evaluated under the same decoder-controlled protocol using frozen encoder features and the lightweight STELLAR decoder. TiTok uses its native larger ViT decoder.

MODEL	# TKS	RECONSTRUCTION				SEMANTICS	
		FID ↓	LPIPS ↓	PSNR ↑	SSIM ↑	LIN.	KNN
DINO	1	-	-	-	-	76.46	74.69
DINO	196	3.27	0.2121	16.36	0.3900	70.31	54.41
MAE	196	3.02	0.2071	16.54	0.3979	66.32	25.82
TiTok*	32	2.75	0.3281	14.97	0.3311	33.42	7.30
TiTok*	64	1.99	0.2571	16.01	0.3721	32.87	7.29
OURS	16	3.06	0.2077	16.57	0.3970	73.26	67.25
OURS	196	2.85	0.2085	16.58	0.3988	72.21	64.71
OURS(H)	16	2.60	0.1729	17.91	0.4682	79.10	77.31

B.2. Pooling Strategy for Sparse Tokens

In the main text, we use mean-pooled sparse tokens for linear probing. To test whether the observed trend over r is an artifact of mean pooling, we additionally evaluate attention pooling and max pooling. As shown in Table 8, $r = 16$ remains the best operating point across pooling strategies, while max pooling performs worse. This suggests that increasing r weakens the semantic bottleneck rather than simply changing the behavior of mean pooling.

Table 8. **Effect of pooling strategy on ImageNet-1K linear probing.** We evaluate sparse-token representations with different numbers of tokens r using attention pooling, mean pooling, and max pooling. The trend remains stable across pooling strategies, suggesting that the decrease in accuracy for larger r is not merely a mean-pooling artifact.

r	ATTENTION POOLING	MEAN POOLING	MAX POOLING
8	72.86	72.97	72.33
16	73.05	73.26	72.17
24	72.15	72.17	70.49

B.3. Sensitivity to Prototype Count

Table 9 studies the effect of the prototype vocabulary size K . Performance is stable for $K \geq 4096$, while $K = 1024$ leads to lower ImageNet and ADE20K performance. We also observed less stable clustering and set-alignment losses at $K = 1024$. This suggests that a sufficiently large prototype vocabulary is helpful for modeling fine-grained visual concepts such as objects, parts, materials, textures, and scene context.

B.4. Effect of Pretraining Data

We pretrained separate STELLAR versions on ImageNet-1K, Places365 (Zhou et al., 2017a) and compared their linear probing performance in Table 10.

Table 9. **Sensitivity to the number of visual prototypes K .** We evaluate ImageNet-1K linear probing and ADE20K segmentation for different prototype vocabulary sizes. Performance is stable once K is sufficiently large, while $K = 1024$ leads to weaker performance.

K	IN1K $\uparrow$	ADE20K $\uparrow$
1024	70.79	30.21
4096	73.12	30.64
8192	73.31	30.98
16384	73.26	31.33

Table 10. Effect of pretraining data.

Pretraining data	linear probing acc.	
	ImageNet-1K	Places365
ImageNet-1K	76.94	49.25
Places365	66.08	51.98

B.5. Dense Prediction from the Low-rank Representation

Table 11 compares segmentation probes trained directly on the low-rank representation LS and on the dense feature map. The low-rank representation retains non-trivial spatial information, especially on Pascal VOC, but it does not fully replace dense features for segmentation. Thus, the dense prediction results in the main text should be interpreted as evidence that sparse factorized training shapes the backbone feature map into a more region-aware representation, rather than as evidence that LS alone is sufficient for all dense prediction tasks.

Table 11. **Segmentation probing with low-rank and dense representations.** We compare linear segmentation probes trained on the low-rank reconstructed representation LS and on the dense feature map from the backbone. The low-rank representation retains non-trivial spatial signal, but dense features remain substantially stronger for dense prediction.

REPRESENTATION	ADE20K $\uparrow$	PASCAL VOC $\uparrow$	CITYSCAPES $\uparrow$
LOW-RANK LS	17.58	70.00	13.60
DENSE FEATURE MAP	31.33	81.83	27.74

B.6. Concept Alignment with Language

Inspired by Zhang et al. (2025), we use frozen STELLAR features and align them with the text tower of CLIP (Radford et al., 2021) using a single attention-pooled probing layer. The evaluation on vision-language tasks, together with comparisons to baseline models, is shown in Table 12. This experiment is intended as a lightweight probing study rather than a fully optimized vision-language training setup.

B.7. Finetuning

We finetune STELLAR on ImageNet-1K classification and ADE20K segmentation and compare it with baseline models. We use the same train/validation splits and hyperparameter selection protocol as in Sec. A.2, but unfreeze the backbone and finetune all model parameters for 75 epochs. All models use ViT-B backbones. As shown in Table 13, STELLAR achieves consistent gains across tasks and performs close to the strongest baseline, iBOT.

B.8. Efficiency Analysis

To analyze the efficiency of STELLAR, we measure the processing time of the main components on a single A100 GPU at different batch sizes. Encoding the main global view takes most of the processing time, followed by encoding the masked views (8 views at 80% masking ratio) and decoding to the original image. The Sinkhorn-Knopp algorithm used for clustering and the Sinkhorn algorithm used for optimal transport matching account for a much smaller fraction of the total runtime, and their processing time remains nearly constant as batch size increases.

We compare our Sinkhorn-based matching procedure with Hungarian matching, which is commonly used in prior work

Table 12. Language alignment evaluation.

	IN-1K 0-shot		MS COCO		Winoground		MMVP
	@1	@5	T2I	I2T	Text	Image	Avg.
MAE	23.18	50.43	11.28	13.46	20.75	9.00	19.26
iBOT	50.01	80.43	20.79	29.38	24.75	12.00	18.52
STELLAR	51.53	80.04	17.94	22.34	26.25	8.25	19.26
CLIP	72.7	-	43.0	59.7	30.5	11.5	20.0

Table 13. Finetuning performance in ImageNet-1K classification accuracy and ADE20K segmentation mIoU (%). We show in parentheses the gain over the respective linear probing results.

Model	ImageNet-1K Acc.	ADE20K mIoU
DINO	79.58 (+3.12)	39.22 (+12.35)
MAE	77.75 (+11.43)	40.33 (+9.42)
iBOT	80.72 (+9.14)	42.76 (+10.97)
STELLAR	80.05 (+6.78)	41.98 (+10.65)

such as Sparse R-CNN (Sun et al., 2021), DETR (Carion et al., 2020), and MaskFormer (Cheng et al., 2021). Since exact Hungarian matching is less amenable to GPU parallelization, its computational cost increases approximately linearly with batch size. At batch size 64, Hungarian matching already costs more than the encoder forward pass, while our Sinkhorn-based matching is over $100\times$ faster. We therefore add a small entropy regularization term to the bipartite matching objective, enabling efficient GPU-parallel matching with Sinkhorn iterations.

Table 14. Processing time (s) of the main components in STELLAR on one A100 GPU at different batch sizes. We compare our Sinkhorn-based matching procedure with Hungarian matching, shown in gray.

Batch size	4	8	16	32	64
Encoder	8.2×10^{-3}	9.1×10^{-3}	1.4×10^{-2}	2.0×10^{-2}	3.2×10^{-2}
Decoder	4.6×10^{-3}	6.8×10^{-3}	8.8×10^{-3}	1.2×10^{-2}	1.5×10^{-2}
Mask encoding	7.9×10^{-3}	8.9×10^{-3}	1.1×10^{-2}	1.8×10^{-2}	1.7×10^{-2}
SK clustering	3.4×10^{-4}	3.4×10^{-4}	3.4×10^{-4}	3.7×10^{-4}	3.9×10^{-4}
Sinkhorn matching	1.4×10^{-3}	1.4×10^{-3}	1.4×10^{-3}	1.4×10^{-3}	1.2×10^{-3}
Hungarian matching	5.7×10^{-3}	1.7×10^{-2}	4.0×10^{-2}	9.0×10^{-2}	1.8×10^{-1}